

Question:

There exist real numbers x and y , both greater than 1, such that $\log_x(y^x)=\log_y(x^{4y})=10$. Find xy .

Reasoning + Answer:

We are given that there exist real numbers x and y , both greater than 1, such that:

$$\log_x(y^x)=10 \quad \text{and} \quad \log_y(x^{4y})=10$$

We are to find xy .

Step 1: Simplify the first equation

$$\log_x(y^x)=10$$

Using logarithmic identities:

$$\log_x(y^x)=x \cdot \log_x y$$

So:

$$x \cdot \log_x y=10 \tag{1}$$

Step 2: Simplify the second equation

$$\log_y(x^{4y})=10$$

Using the identity $\log_y(a^b)=b \log_y a$:

$$4y \cdot \log_y x=10 \tag{2}$$

Now, from equations (1) and (2):

From (1):

$$x \cdot \log_x y=10 \Rightarrow \log_x y=\frac{10}{x}$$

From (2):

$$4y \cdot \log_y x=10 \Rightarrow \log_y x=\frac{10}{4y}=\frac{5}{2y}$$

Step 3: Use the identity $\log_a b=\frac{1}{\log_b a}$

We know that:

$$\log_x y=\frac{1}{\log_y x}$$

So substitute the expressions:

$$\text{Left: } \log_x y=\frac{10}{x}$$

$$\text{Right: } \frac{1}{\log_y x}=\frac{1}{5/(2y)}=\frac{2y}{5}$$

Therefore:

$$\frac{10}{x}=\frac{2y}{5}$$

Multiply both sides by x :

$$10=\frac{2y}{5} \cdot x \Rightarrow 10=\frac{2xy}{5}$$

Multiply both sides by 5:

$$50=2xy \Rightarrow xy=\frac{50}{2}=25$$

<answer>25</answer>